

Roof Opening Panel



Special Tool(s)

 ST2574-A	Flex Probe Kit NUD105-R025D or equivalent
 ST3093-A	Fluke 77-IV Digital Multimeter FLU77-4 or equivalent

Principles of Operation

The roof opening panel assembly uses an integrated motor and module to operate the roof opening panel. The roof opening panel motor and module are installed new as an assembly. The roof opening panel is an electronically operated glass panel that can be opened, closed or vented by the roof opening panel switch.

The roof opening panel motor must be initialized when repairs are carried out on any part of the roof opening panel system, including whenever:

- the roof opening panel motor has been removed from the roof opening panel system.
- the roof opening panel glass has been removed from the roof opening panel system.
- the roof opening panel assembly has been removed from the vehicle.
- a new roof opening panel motor has been installed.
- a new roof opening panel glass has been installed.
- a new roof opening panel assembly has been installed.

Inspection and Verification

1. Verify the customer concern.
2. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection Chart

Mechanical	Electrical
• Roof opening panel • Roof opening panel frame • Roof opening panel drain hoses • Roof opening panel lifter assemblies • Roof opening panel water trough • Roof opening panel air deflector assembly • Roof opening panel weatherstrip • Roof opening panel shield	• Battery Junction Box (BJB) fuse 48 (20A) • Body Control Module (BCM) fuse 32 (15A) • Wiring, terminals or connectors • Accessory delay relay • Overhead console switch

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.
4. If the concern is not visually evident, determine the symptom. GO to [Symptom Chart — Roof Opening Panel](#) or GO to [Symptom Chart — NVH](#).

Symptom Chart — Roof Opening Panel

Symptom Chart — Roof Opening Panel		
Condition	Possible Causes	Action
• The roof opening panel leaks	• Roof opening panel alignment • Roof opening panel weatherstrip • Roof opening panel rear trough guide • Damaged, clogged or obstructed roof opening panel drain hoses • Roof opening panel frame • Incorrect position of the roof opening panel glass • Sealant joints at frame (rail and track assembly) joints	• GO to Pinpoint Test B.
• The roof opening panel does not open or close	• Fuse(s) • Accessory delay relay • Wiring, terminals or connectors	• GO to Pinpoint Test E.

	• Overhead console switch • Roof opening panel motor/module assembly	
• The roof opening panel does not open or close in VENT position	• Wiring, terminals or connectors • Overhead console switch • Roof opening panel motor/module assembly	• GO to Pinpoint Test F.
• The express open/close is inoperative	• Roof opening panel motor not initialized • Roof opening panel motor/module	• INITIALIZE the roof opening panel motor/module. REFER to Roof Opening Panel Motor Initialization in this section. If concern is still present after initialization, INSTALL a new roof opening panel motor/module. REFER to Roof Opening Panel Motor in this section. TEST the system for normal operation.
• The roof opening panel does not stop in FLUSH from any position	• Incorrect roof opening panel alignment	• ALIGN the roof opening panel. REFER to Roof Opening Panel Alignment in this section.
• Roof opening panel reverses during operation	• Incorrect roof opening panel alignment • Foreign material in guide track(s)	• ALIGN the roof opening panel. REFER to Roof Opening Panel Alignment in this section. • REMOVE foreign material from guide track.

Symptom Chart — NVH

Symptom Chart — NVH		
Condition	Possible Causes	Action
NOTE: NVH symptoms should be identified using the diagnostic tools that are available. For a list of these tools, an explanation of their uses and a glossary of common terms, refer to Section 100-04. Since it is possible any one of multiple systems may be the cause of a symptom, it may be necessary to use a process of elimination type of diagnostic approach to pinpoint the responsible system. If this is not the causal system for the symptom, refer back to Section 100-04 for the next likely system and continue diagnosis.		
• The roof opening panel has excessive wind noise	• Roof opening panel glass alignment • Roof opening panel weatherstrip • Roof opening panel air deflector assembly	• GO to Pinpoint Test A.

• The roof opening panel rattles	• Roof opening panel glass alignment • Roof opening panel air deflector assembly • Roof opening panel water trough and guides • Roof opening panel lifter assemblies • Roof opening panel tracks • Roof opening panel shield (sunshade) • Roof opening panel assembly	• GO to Pinpoint Test C.
• The roof opening panel is noisy during operation	• Obstructed or damaged roof opening panel • Roof opening panel lifter assemblies • Roof opening panel alignment • Noisy roof opening panel motor/module assembly • Roof opening panel shield (sunshade)	• GO to Pinpoint Test D.

Pinpoint Tests

Pinpoint Test A: The Roof Opening Panel Has Excessive Wind Noise

Normal Operation

When the roof opening panel glass is completely closed, it is flush with the roof panel.

This pinpoint test is intended to diagnose the following:

- Roof opening panel glass alignment
- Roof opening panel weatherstrip
- Roof opening panel air deflector assembly

PINPOINT TEST A : THE ROOF OPENING PANEL HAS EXCESSIVE WIND NOISE

A1 VERIFY ROOF OPENING PANEL NOISE	
• Road test the vehicle with the roof opening panel in the CLOSED and OPEN position.	
Is the wind noise present only in the OPEN position?	
Yes	GO to A4 .
No	GO to A2 .
A2 CHECK THE ROOF OPENING PANEL ALIGNMENT	

- Make sure the roof opening panel glass is centered and flush with the roofline.

Is the roof opening panel aligned correctly?

Yes	GO to A3 .
No	ALIGN the roof opening panel. REFER to Roof Opening Panel Alignment in this section. TEST the system for normal operation.

A3 CHECK THE ROOF OPENING PANEL WEATHERSTRIP

NOTE: Make sure the weatherstrip is secured to the glass, not cut, cracked, pinched, loose or obstructed in CLOSED, VENT and OPEN positions.

- Check the roof opening panel weatherstrip.

Is the roof opening panel weatherstrip OK?

Yes	REFER to Section 501-00 to diagnose other possible sources of the excessive wind noise.
No	INSTALL a new roof opening panel weatherstrip. TEST the system for normal operation.

A4 CHECK THE AIR DEFLECTOR

- Make sure the air deflector is securely attached and check for torn and unsecured tethers.

Is the air deflector OK?

Yes	GO to A5 .
No	REPAIR or INSTALL a new air deflector assembly. REFER to Air Deflector in this section. TEST the system for normal operation.

A5 CHECK THE AIR DEFLECTOR FOR VIBRATION

- Road test the vehicle and check for air deflector vibration.

Did the air deflector vibrate?

Yes	MODIFY the curvature of the air deflector and RETEST.
No	PARTIALLY close the roof opening panel until the noise is eliminated. If the noise is eliminated, ADVISE the customer that changing glass location eliminates the wind noise. If the wind noise is not eliminated, INSTALL a new air deflector assembly. REFER to Air Deflector in this section. TEST the system for normal operation.

Pinpoint Test B: The Roof Opening Panel Leaks

Normal Operation

When the roof opening panel glass is completely closed, it is flush with the roof panel.

This pinpoint test is intended to diagnose the following:

- Roof opening panel alignment
- Roof opening panel weatherstrip
- Roof opening panel rear trough guide
- Damaged, clogged or obstructed roof opening panel drain hoses
- Roof opening panel frame
- Incorrect position of the roof opening panel glass
- Sealant joints at frame (rail and track assembly) joints

PINPOINT TEST B : THE ROOF OPENING PANEL LEAKS	
B1 VERIFY CUSTOMER OPERATION OF THE ROOF OPENING PANEL WHEN PARKED	
• Make sure the customer is parking with the roof opening panel in the CLOSED position.	
Is the customer parking with the roof opening panel in the CLOSED position?	
Yes	GO to B2 .
No	ADVISE the customer to refer to the Owner's Literature for correct operation of the roof opening panel.
B2 CHECK THE ALIGNMENT OF THE ROOF OPENING PANEL	
• Make sure the roof opening panel glass is centered and flush with the roof line.	
Is the roof opening panel aligned correctly?	
Yes	GO to B3 .
No	ALIGN the roof opening panel. REFER to Roof Opening Panel Alignment in this section. TEST the system for normal operation.
B3 CHECK THE WEATHERSTRIP CONDITION	
NOTE: <i>The weatherstrip has a bonded joint located on the passenger side or rear edge. This is not a defect.</i>	
• Make sure the roof opening panel weatherstrip is not cracked, torn or loose. • Check the roof opening panel weatherstrip.	
Is the roof opening panel weatherstrip in good condition and free from damage?	
Yes	GO to B4 .
No	INSTALL a new roof opening panel weatherstrip. TEST the system for normal operation.
B4 CHECK THE WATER TROUGH AND GUIDE CONDITION	
• Check the water trough and guide for damage and/or incorrect attachment and function.	

Is the water trough and guide OK?

Yes	GO to B5 .
No	REPAIR the water trough or INSTALL a new component as necessary. TEST the system for normal operation.

B5 CHECK THE DRAIN HOSES

NOTE: *Drain hoses can be viewed with a flashlight by removing the headliner lace around the inside opening and gently pulling down on the headliner. (On some vehicles, hook-and-loop tape style attachments have been installed in place of trim lace.)*

- Make sure the drain hoses are correctly attached, not cracked, slit, pinched or obstructed.

Are all the drain hoses present and attached correctly?

Yes	GO to B6 .
No	REPAIR or INSTALL new drain hose(s) as necessary. TEST the system for normal operation.

B6 CHECK THE DRAIN HOSES FOR CORRECT FLOW

NOTE: *Drain hoses must flow downhill and cannot be pinched or restricted by headliner or any other trim components. For drain hose routing information, refer to Roof Opening Panel in this section.*

- With the panel OPEN, slowly pour water into the water trough area and make sure that the same amount of water exits the vehicle at the bottom of the A- and C-pillars.
- Check the drain hoses for obstructions by blowing compressed air through the drain spouts in the water trough area, then retest with water.

Are the drain hoses operating correctly?

Yes	INSPECT other areas of the vehicle that may cause water to enter the roof area such as the windshield. If no concern is found, GO to B7 .
No	LOWER the headliner and INSPECT the drain hoses for correct routing. REPAIR or INSTALL a new drain hose(s). TEST the system for normal operation.

B7 CHECK THE CONDITION OF THE ROOF OPENING PANEL FRAME

- Check the roof opening panel frame for damage that may cause the roof opening panel seal to seat incorrectly.

Is the roof opening panel frame OK?

Yes	INSTALL a new roof opening panel. REFER to Roof Opening Panel in this section.
No	REPAIR roof opening panel frame as necessary. TEST the system for normal operation.

Pinpoint Test C: The Roof Opening Panel Rattles

Normal Operation

When the roof opening panel glass is completely closed, it is flush with the roof panel. All components should be secured correctly.

This pinpoint test is intended to diagnose the following:

- Roof opening panel glass alignment
- Roof opening panel air deflector assembly
- Roof opening panel water trough and guides
- Roof opening panel lifter assemblies
- Roof opening panel tracks
- Roof opening panel shield (sunshade)
- Roof opening panel assembly

PINPOINT TEST C : THE ROOF OPENING PANEL RATTLES

C1 CHECK THE ROOF OPENING PANEL TIGHTNESS	
• With the roof opening glass panel in the CLOSED position, manually push up and fore/aft on the glass panel.	
Is the roof opening panel glass loose?	
Yes	ALIGN the roof opening panel. REFER to Roof Opening Panel Alignment in this section. TEST the system for normal operation.
No	GO to C2 .
C2 CHECK THE ROOF OPENING PANEL TRACKS	
• Open the roof opening panel. • Check the roof opening panel tracks for obstructions or foreign material and damage.	
Are the roof opening panel tracks OK?	
Yes	GO to C3 .
No	REMOVE all the roof opening panel track obstructions and REPAIR the roof opening panel tracks. TEST the system for normal operation.
C3 CHECK THE ROOF OPENING PANEL WATER TROUGH AND REAR TROUGH GUIDES	
• Check the roof opening panel water trough and rear trough guides for any obstructions.	
Are the roof opening panel water trough and rear trough guides installed securely and free from obstructions?	
Yes	GO to C4 .
No	REMOVE all obstructions from the roof opening panel water trough. SECURE the trough guides or INSTALL new roof opening panel rear trough guides if damaged. REFER to Roof Opening Panel Rear Trough Guide in this section. TEST the system for normal operation.

C4 CHECK THE ROOF OPENING PANEL SHIELD (SUNSHADE)

- Check the roof opening panel shield (sunshade) for correct installation with all tabs in track channels and the guide feet intact.

Is the roof opening panel shield installed correctly?

Yes	GO to C5 .
No	INSTALL the roof opening panel shield correctly. REFER to Roof Opening Panel Shield in this section. TEST the system for normal operation.

C5 CHECK THE ROOF OPENING PANEL AIR DEFLECTOR ASSEMBLY

- Check the roof opening panel air deflector assembly for obstructions, damage and incorrect attachments.

Is the roof opening panel air deflector assembly OK?

Yes	GO to C6 .
No	REMOVE all obstructions and REPAIR or INSTALL a new roof opening panel air deflector assembly. REFER to Air Deflector in this section. TEST the system for normal operation.

C6 CHECK THE ROOF OPENING PANEL LIFTER ASSEMBLIES

- Verify the roof opening panel lifter assemblies are free from obstructions and damage.

Are the roof opening panel lifter assemblies OK?

Yes	GO to C7 .
No	REMOVE all obstructions from the roof opening panel lifter assemblies and REPAIR any damage. TEST the system for normal operation.

C7 CHECK THE ROOF OPENING PANEL FRAME ATTACHMENTS

- Check for correct torque and installation of the roof opening panel frame.

Is the roof opening panel correctly installed and tightened to specification?

Yes	GO to C8 .
No	TIGHTEN or REINSTALL the roof opening panel frame. REFER to Roof Opening Panel Frame in this section. TEST the system for normal operation.

C8 CHECK THE ROOF OPENING PANEL FRAME TO HEADLINER FOR RATTLES

- Check roof opening panel frame to headliner for any interference condition that would induce a rattle.

Is there a condition that causes a rattle?

Yes	INSTALL a suitable sound damper material to correct the condition. If the rattle still persists, INSTALL a new roof opening panel frame or headliner. REFER to Roof Opening Panel Frame in this section. TEST the system for normal operation.
No	INSTALL a new roof opening panel frame. REFER to Roof Opening Panel Frame in this section. TEST the system for normal operation.

Pinpoint Test D: The Roof Opening Panel is Noisy During Operation

Normal Operation

The roof opening panel operates smoothly while opening and closing.

This pinpoint test is intended to diagnose the following:

- Obstructed or damaged roof opening panel
- Roof opening panel lifter assemblies
- Roof opening panel alignment
- Noisy roof opening panel motor/module assembly
- Roof opening panel shield (sunshade)

PINPOINT TEST D : THE ROOF OPENING PANEL IS NOISY DURING OPERATION

D1 CHECK THE ROOF OPENING PANEL FOR OBSTRUCTIONS OR DAMAGE	
• Check the roof opening panel for any obstructions and damage.	
Are there any obstructions and/or damage?	
Yes	REMOVE all obstructions. REPAIR any damage or INSTALL a new roof opening panel frame. REFER to Roof Opening Panel Frame in this section. TEST the system for normal operation.
No	GO to D2 .
D2 CHECK THE ROOF OPENING PANEL ALIGNMENT	
• Check the roof opening panel during operation.	
Is the roof opening panel loose or incorrectly aligned?	
Yes	ALIGN the roof opening panel. REFER to Roof Opening Panel Alignment in this section. TEST the system for normal operation.
No	GO to D3 .
D3 CHECK THE ROOF OPENING PANEL TRACKS	
• Open the roof. • Check the roof opening panel tracks for obstructions and damage.	

Are the roof opening panel tracks OK?

Yes	GO to D4 .
No	REMOVE all obstructions and REPAIR any damage. TEST the system for normal operation.

D4 CHECK THE ROOF OPENING PANEL SHIELD (SUNSHADE)

- Check the roof opening panel shield (sunshade) for correct movement.

Is the roof opening panel shield moving correctly?

Yes	GO to D5 .
No	REMOVE and REINSTALL the roof opening panel shield. REFER to Roof Opening Panel Shield in this section. TEST the system for normal operation.

D5 CHECK THE ROOF OPENING PANEL LIFTER ASSEMBLIES

- Check the roof opening panel lifter assemblies for obstructions and damage.

Are the roof opening panel lifter assemblies OK?

Yes	GO to D6 .
No	REMOVE all obstructions and REPAIR any damage. TEST the system for normal operation.

D6 CHECK THE ROOF OPENING PANEL MOTOR/MODULE

- Remove the roof opening panel motor/module and leave the electrical connector connected.
- Operate the roof opening panel to the full OPEN and CLOSED position.

Does the motor/module make excessive noise?

Yes	INSTALL a new roof opening panel motor/module. REFER to Roof Opening Panel Motor in this section. TEST the system for normal operation.
No	REINSTALL the roof opening panel motor. REFER to Roof Opening Panel Motor in this section. TEST the system for normal operation. If the roof opening panel is still noisy, INSTALL a new roof opening panel frame. REFER to Roof Opening Panel Frame in this section. If the roof opening panel is no longer noisy, the system is operating correctly. The concern may have been caused by an incorrectly installed roof opening panel motor.

Pinpoint Test E: The Roof Opening Panel Does Not Open or Close

Refer to Wiring Diagrams Cell [101](#) , Roof Opening Panel for schematic and connector information.

The roof opening panel motor must be initialized when repairs are carried out on any part of the roof opening panel system, including whenever:

- the roof opening panel motor has been removed from the roof opening panel system.
- the roof opening panel glass has been removed from the roof opening panel system.
- the roof opening panel assembly has been removed from the vehicle.
- a new roof opening panel motor has been installed.
- a new roof opening panel glass has been installed.
- a new roof opening panel assembly has been installed.

Refer to Roof Opening Panel Motor Initialization in this section.

Normal Operation

The roof opening panel uses an integrated motor and module to operate the roof opening panel. A new roof opening panel motor/module is installed as an assembly.

The roof opening panel motor/module receives voltage from the accessory delay relay when the ignition is in the ACC or RUN position. The overhead console switch supplies ground signals to the roof opening panel motor/module to operate the roof opening panel.

This pinpoint test is intended to diagnose the following:

- Fuse(s)
- Accessory delay relay
- Wiring, terminals or connectors
- Overhead console switch
- Roof opening panel motor/module assembly

PINPOINT TEST E : THE ROOF OPENING PANEL DOES NOT OPEN OR CLOSE

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

E1 CHECK THE POWER WINDOWS

- Ignition ON.
- Verify the power windows operate.

Do the power windows operate?

Yes	GO to E2 .
No	REFER to Section 501-11 to diagnose the accessory delay relay circuit.

E2 CHECK THE MOTOR/MODULE FOR INITIALIZATION

- Attempt to carry out the Roof Opening Panel Motor Initialization procedure. Refer to Roof Opening Panel Motor Initialization in this section.

Is the concern still present, or does the roof opening panel motor fail to initialize?

Yes	GO to E3 .
No	The system is operating normally at this time. The roof opening panel motor was not initialized.

E3 CHECK THE OVERHEAD CONSOLE SWITCH

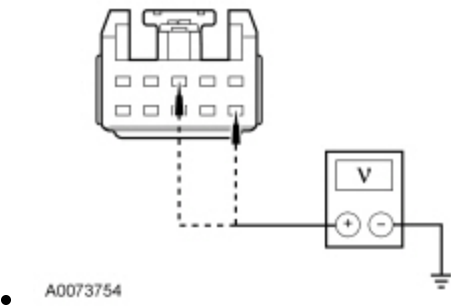
- Ignition OFF.
- Disconnect: Overhead Console Switch [C912](#).
- Carry out the overhead console switch component test.
Refer to Wiring Diagrams Cell [149](#) for component testing.

Is the overhead console switch OK?

Yes	GO to E4 .
No	INSTALL a new overhead console switch. REFER to Roof Opening Panel Control Switch in this section. TEST the system for normal operation.

E4 CHECK VOLTAGE TO THE ROOF OPENING PANEL MOTOR/MODULE

- Disconnect: Roof Opening Panel Motor/Module [C921](#).
- Ignition ON.
- Measure the voltage between roof opening panel motor/module [C921](#) Pin 3, circuit CBP32 (GN/VT), harness side and roof opening panel motor/module [C921](#) Pin 6, circuit SBB48 (VT/RD), harness side and ground.

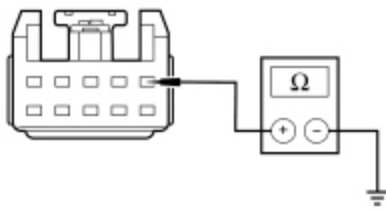


Are the voltages greater than 10 volts for both measurements?

Yes	GO to E5 .
No	VERIFY Battery Junction Box (BJB) fuse 48 (20A) and BCM fuse 32 (15A) are OK. If OK, REPAIR circuit SBB48 (VT/RD) and/or CBP32 (GN/VT) for an open. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. TEST the system for normal operation.

E5 CHECK THE ROOF OPENING PANEL MOTOR/MODULE GROUND

- Ignition OFF.
- Measure the resistance between roof opening panel motor/module [C921](#) Pin 1, circuit GD139 (BK/YE), harness side and ground.



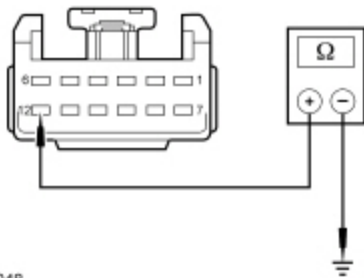
• A0073757

Is the resistance less than 5 ohms?

Yes	GO to E6 .
No	REPAIR the circuit. TEST the system for normal operation.

E6 CHECK THE OVERHEAD CONSOLE SWITCH GROUND

- Measure the resistance between overhead console switch [C912](#) Pin 12, circuit CPR39 (VT/WH), harness side and ground.



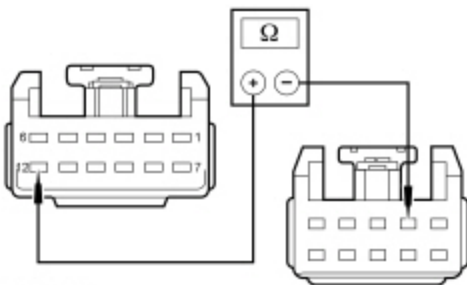
• N0092048

Is the resistance less than 5 ohms?

Yes	GO to E8 .
No	GO to E7 .

E7 CHECK THE COMMON CIRCUIT FOR AN OPEN

- Measure the resistance between overhead console switch [C912](#) Pin 12, circuit CPR39 (VT/WH), harness side and roof opening panel motor/module [C921](#) Pin 2, circuit CPR39 (VT/WH), harness side.



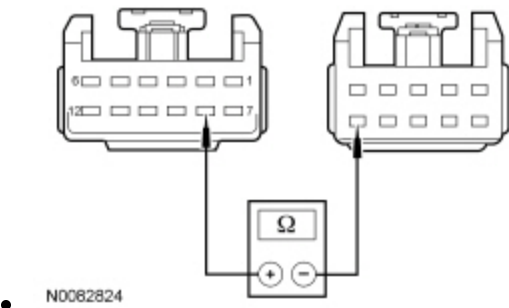
• N00109006

Is the resistance less than 5 ohms?

Yes	INSTALL a new roof opening panel motor/module. REFER to Roof Opening Panel Motor in this section. TEST the system for normal operation.
No	REPAIR the circuit. TEST the system for normal operation.

E8 CHECK THE ROOF OPENING PANEL CONTROL CIRCUIT FOR AN OPEN

- Measure the resistance between overhead console switch [C912](#) Pin 8, circuit CPR40 (YE/OG), harness side and roof opening panel motor/module [C921](#) Pin 10, circuit CPR40 (YE/OG), harness side.

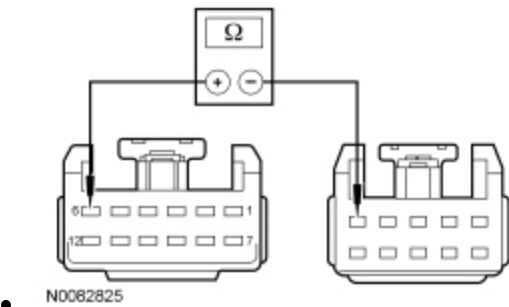


Is the resistance less than 5 ohms?

Yes	GO to E9 .
No	REPAIR the circuit. TEST the system for normal operation.

E9 CHECK THE ROOF OPENING PANEL CONTROL CIRCUIT FOR AN OPEN

- Measure the resistance between overhead console switch [C912](#) Pin 6, circuit CPR31 (VT/BN), harness side and roof opening panel motor/module [C921](#) Pin 5, circuit CPR31 (VT/BN), harness side.



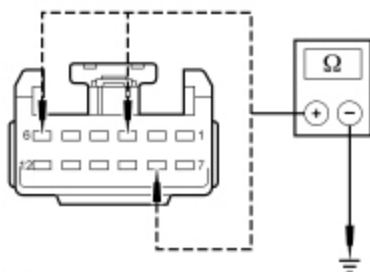
Is the resistance less than 5 ohms?

Yes	GO to E10 .
No	REPAIR the circuit. TEST the system for normal operation.

E10 CHECK THE ROOF OPENING PANEL CONTROL CIRCUITS FOR A SHORT TO GROUND

- Measure the resistance between overhead console switch [C912](#), harness side and ground using the following chart:

Overhead Console Switch	Circuit
C912 Pin 6	CPR31 (VT/BN)
C912 Pin 3	CPR38 (WH/BU)
C912 Pin 8	CPR40 (YE/OG)



N0092049

Is the resistance greater than 10,000 ohms for all measurements?

Yes	GO to E11 .
No	REPAIR the circuit(s). TEST the system for normal operation.

E11 CHECK THE ROOF OPENING PANEL MOTOR/MODULE FOR CORRECT OPERATION

- Disconnect all roof opening panel motor/module connectors.
- Check for:
 - corrosion.
 - bent pins.
 - pushed-out pins.
- Connect all roof opening panel motor/module connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new roof opening panel motor/module. REFER to Roof Opening Panel Motor in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. TEST the system for normal operation.

Pinpoint Test F: The Roof Opening Panel Does Not Open or Close in VENT Position

Refer to Wiring Diagrams Cell [101](#) , Roof Opening Panel for schematic and connector information.

The roof opening panel motor must be initialized when repairs are carried out on any part of the roof opening panel system, including whenever:

- the roof opening panel motor has been removed from the roof opening panel system.
- the roof opening panel glass has been removed from the roof opening panel system.
- the roof opening panel assembly has been removed from the vehicle.
- a new roof opening panel motor has been installed.
- a new roof opening panel glass has been installed.
- a new roof opening panel assembly has been installed.

Refer to Roof Opening Panel Motor Initialization in this section.

Normal Operation

The roof opening panel uses an integrated motor and module to operate the roof opening panel. A new roof opening panel motor/module is installed as an assembly.

The roof opening panel motor/module receives voltage from the accessory delay relay when the ignition is in the ACC or RUN position. The overhead console switch supplies ground signals to the roof opening panel motor/module to operate the roof opening panel.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Overhead console switch
- Roof opening panel motor/module assembly

PINPOINT TEST F : THE ROOF OPENING PANEL DOES NOT OPEN OR CLOSE IN VENT POSITION

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

F1 CHECK THE ROOF OPENING PANEL MOTOR/MODULE FOR INITIALIZATION

- Attempt to carry out the roof opening panel motor initialization procedure. Refer to Roof Opening Panel Motor Initialization in this section.

Is the concern still present, or does the roof opening panel motor fail to initialize?

Yes	GO to F2 .
No	The system is operating normally at this time. The roof opening panel motor was not initialized.

F2 CHECK THE OVERHEAD CONSOLE SWITCH

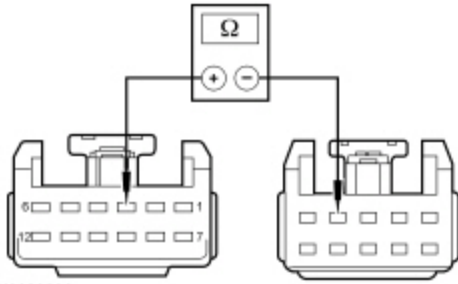
- Ignition OFF.
- Disconnect: Overhead Console Switch [C912](#).
- Carry out the overhead console switch component test.
Refer to Wiring Diagrams Cell [149](#) for component testing.

Is the overhead console switch OK?

Yes	GO to F3 .
No	INSTALL a new roof opening panel switch. REFER to Roof Opening Panel Control Switch in this section. TEST the system for normal operation.

F3 CHECK THE ROOF OPENING PANEL CONTROL CIRCUIT FOR AN OPEN

- Measure the resistance between overhead console switch [C912](#) Pin 3, circuit CPR38 (WH/BU), harness side and roof opening panel motor/module [C921](#) Pin 4, circuit CPR38 (WH/BU), harness side.



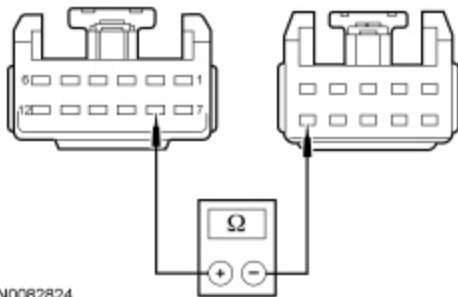
• N0082825

Is the resistance less than 5 ohms?

Yes	GO to F4 .
No	REPAIR the circuit. TEST the system for normal operation.

F4 CHECK THE ROOF OPENING PANEL CONTROL CIRCUIT FOR AN OPEN

- Measure the resistance between overhead console switch [C912](#) Pin 8, circuit CPR40 (YE/OG), harness side and roof opening panel motor/module [C921](#) Pin 10, circuit CPR40 (YE/OG), harness side.



• N0082824

Is the resistance less than 5 ohms?

Yes	GO to F5 .
No	REPAIR the circuit. TEST the system for normal operation.

F5 CHECK THE ROOF OPENING PANEL MOTOR/MODULE FOR CORRECT OPERATION

- Disconnect all roof opening panel motor/module connectors.
- Check for:
 - corrosion.
 - bent pins.
 - pushed-out pins.
- Connect all roof opening panel motor/module connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new roof opening panel motor/module. REFER to Roof Opening Panel Motor in this section. TEST the system for normal operation.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. TEST the system for normal operation.